Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*	(i)		Both distances / 20 cm **and** 1.5 m / <u>drop</u> height	1			1		1
		(ii)		Reduces effect of anomalies or random errors / identify anomalies (1) Accept outliers. Don't accept no anomalies So the <u>mean</u> is more <u>accurate</u> (1)	2			2		2
		(iii)		There are variations in the mass of each case (1) So weighing gives a more accurate value or results (1) **OR** Variations in the mass of each case is very small (1) Using the mean will not affect the results (1)		2		2		2
	(b)	(i)		As case <u>accelerates</u> the air resistance increases (1) until air resistance balances the weight of the case[s] / until the resultant force is zero / forces become balanced (1) Don't accept forces are equal or gravity	2			2		2
		(ii)		Measure split times e.g. over each 50 cm or 75 cm (1) All [split] times should be equal (1) **OR** Drop from >20 cm above pointer / drop from higher point (1) Check whether it affects the times / check the results are the same (1) **OR** Use multi-flash photographs / strobe (1) Check whether [successive] distances are equal (1)			2	2		2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		Substitution: $2.54 = \frac{1.5}{t}$ (1) Manipulation so $t = \frac{1.5}{2.54}$ = 0.59 [s] (1) Accept 0.6 [s] Don't award marks for incorrect substitution	1	1		2	2	2
		(ii)		Mass of 250 cases = 100 g (1) Mass of 1 case = 0.4 g (1) Substitution and conversion: $\left(\frac{4 \text{ x } 0.4\textbf{ecf}}{1000}\right) \text{x } 10$ (1) Air resistance = 0.016 [N] (1) Answer of 16 N **or** 0.004 [N] award 3 marks Answer of 4 N award 2 marks **Alternative:** Mass of 4 cases = 1.6 g (2) = 1.6×10^{-3} kg **ecf** (1) Air resistance = 0.016 [N] (1)		4		4	3	4
				Question 4 total	**6**	**7**	**2**	**15**	**5**	**15**